

Ethanol Blending in Gasoline – France

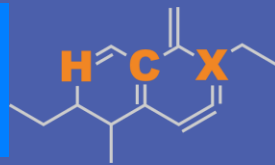
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Ethanol Blending in Europe

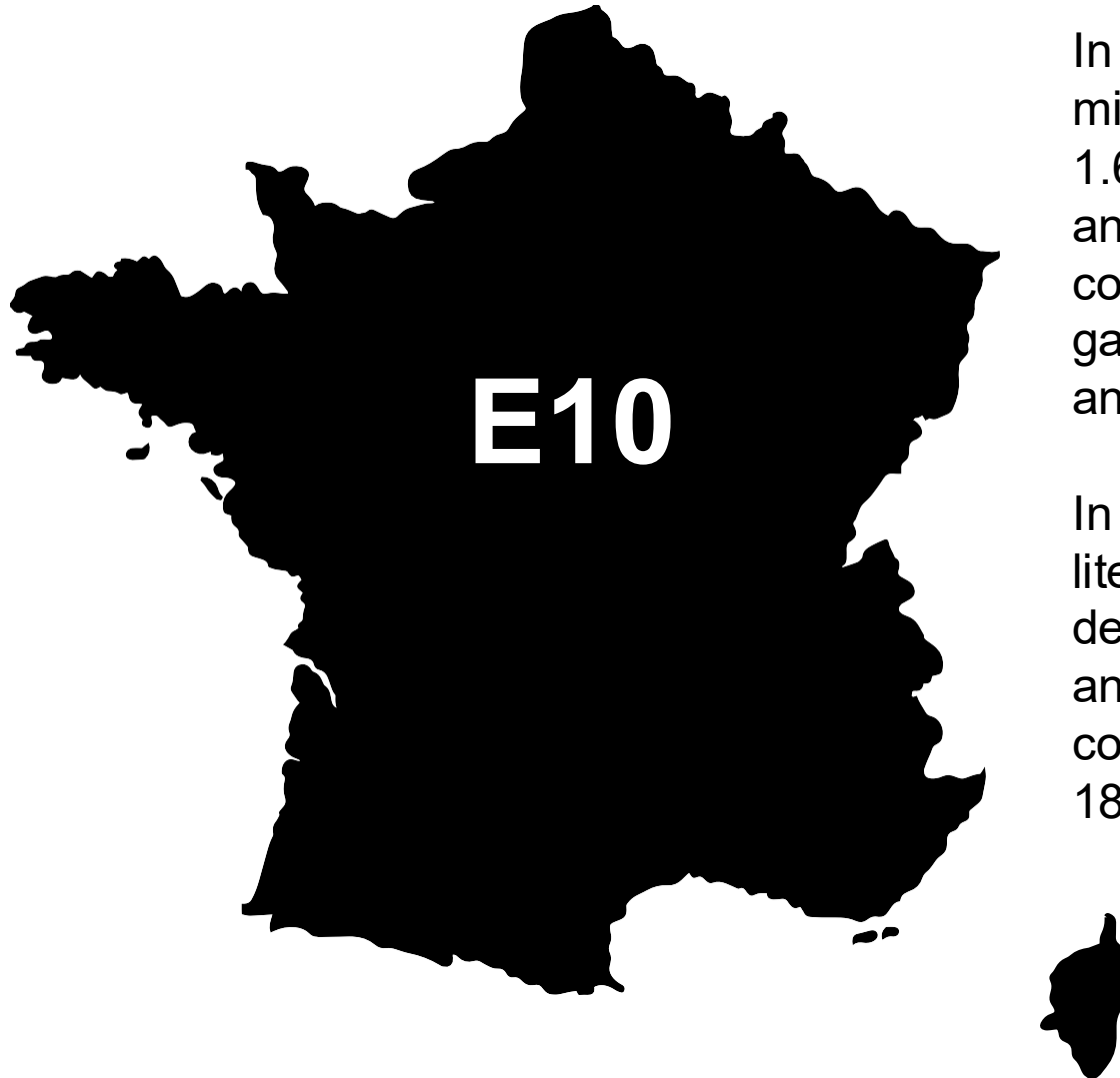
There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in France

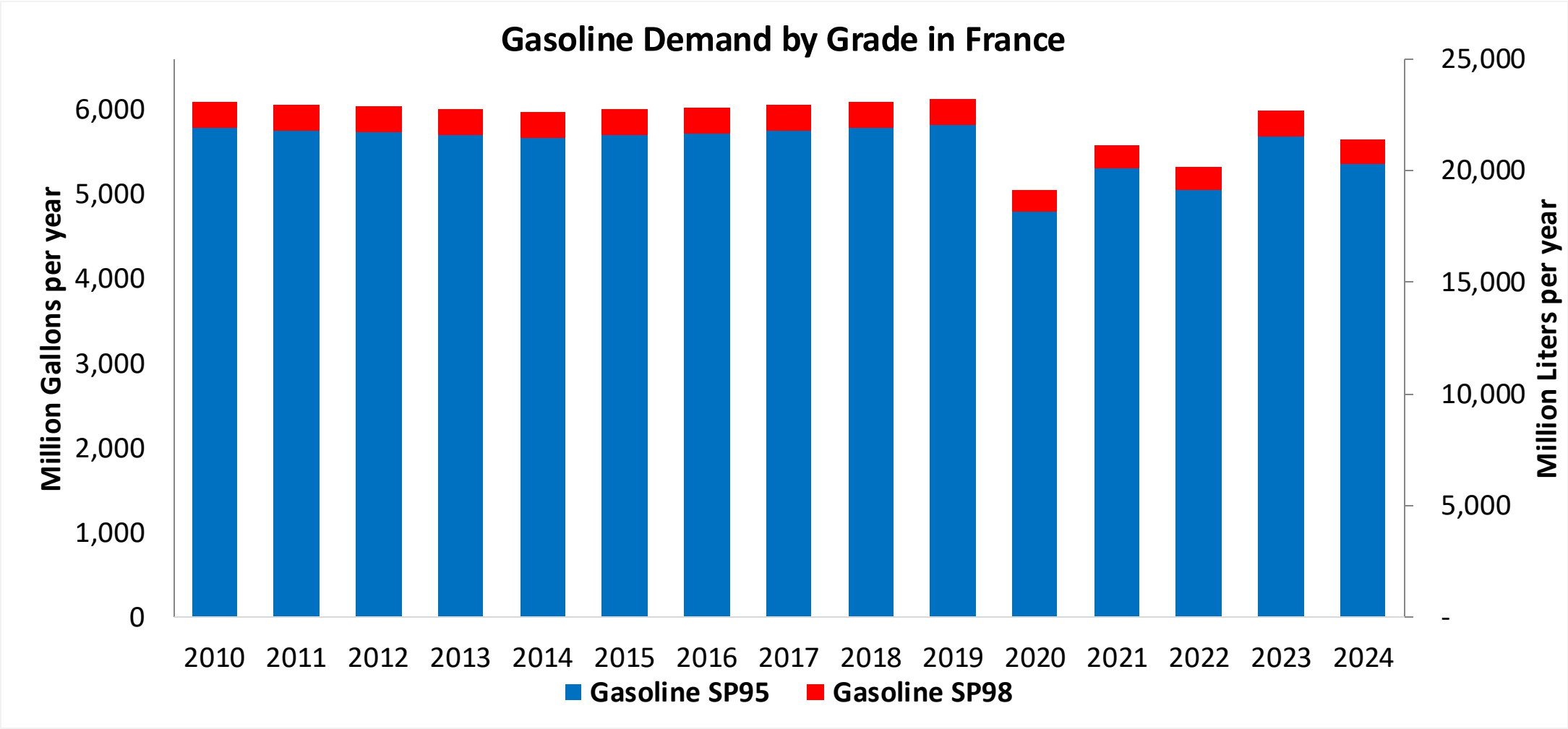


In 2024, gasoline consumption in France was 21,379 million liters (5,648 million gallons) and growth rate was - 1.6%. Gasoline production and net imports were 10,333 and 11,046 million liters (2,730 and 2,918 million gallons) correspondingly. France complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.2 RON.

In 2024, ethanol consumption in France was 2,622 million liters (693 million gallons) representing 12.3% of gasoline demand. Ethanol production and Net Imports were 2,154 and 468 million liters (693 and 124 million gallons) correspondingly. Ethanol sources are Corn 27%, Cereals 18% and Sugar Beet 37%.

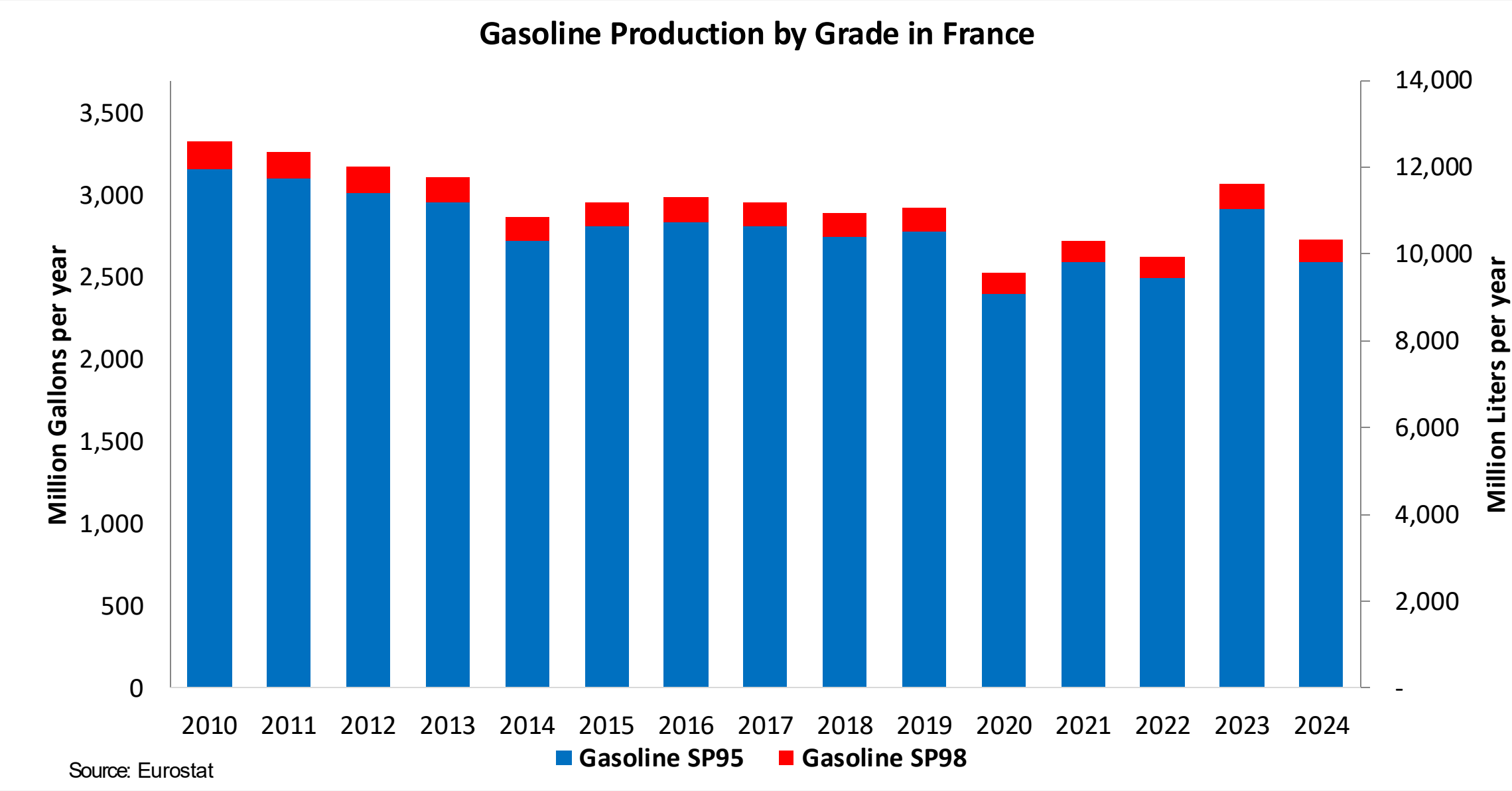
Source: Eurostat, European Commission

Gasoline Demand in France

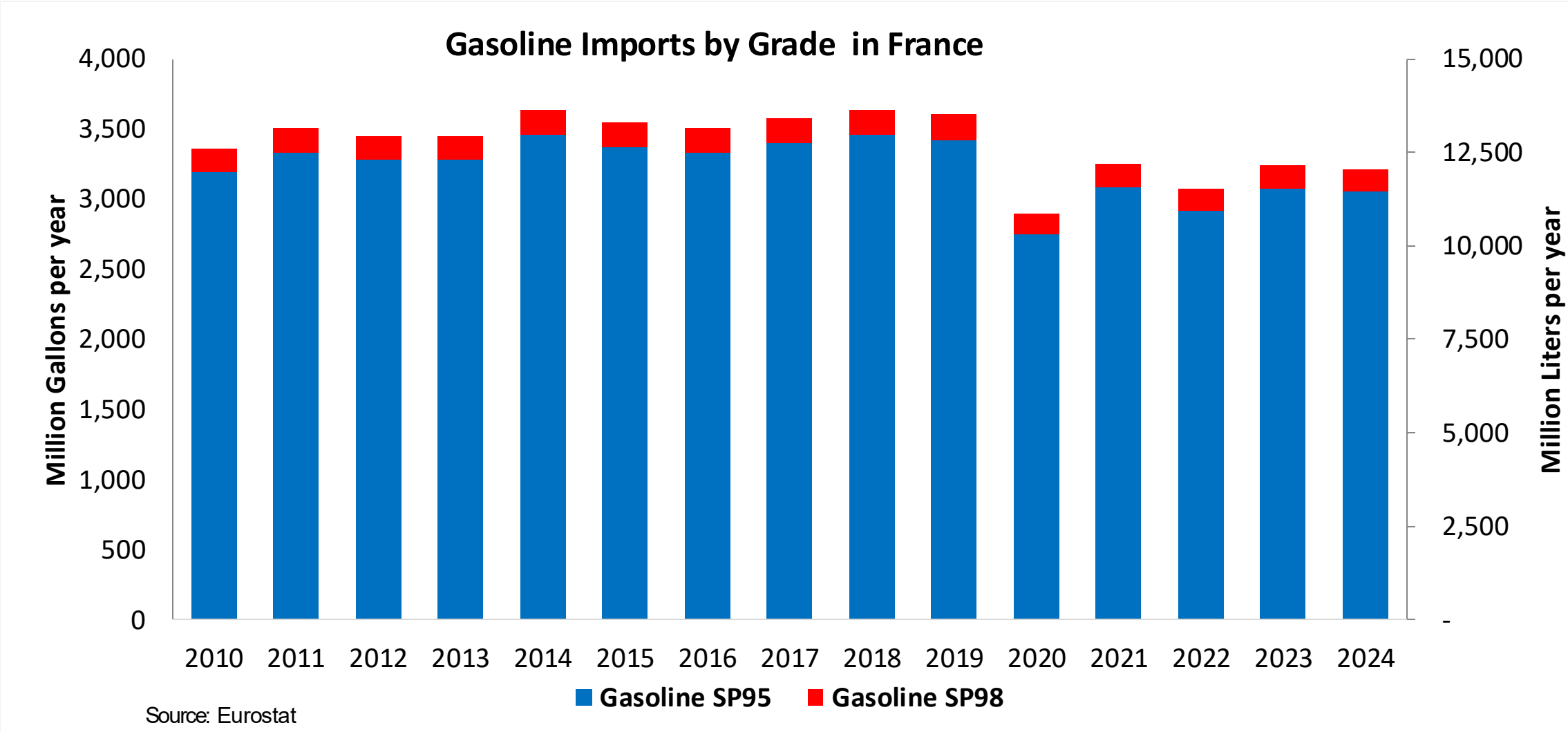


Source: Eurostat

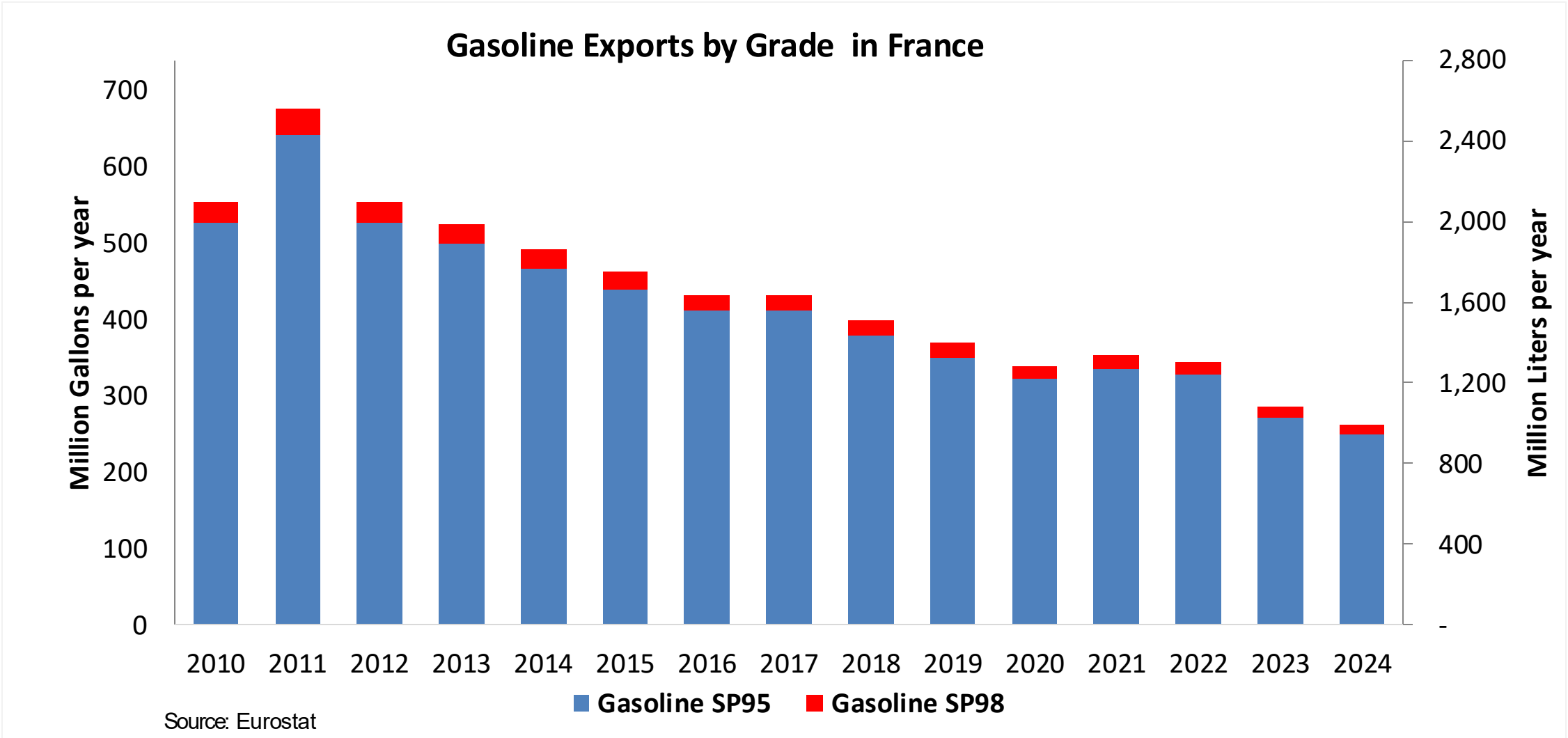
Gasoline Production in France



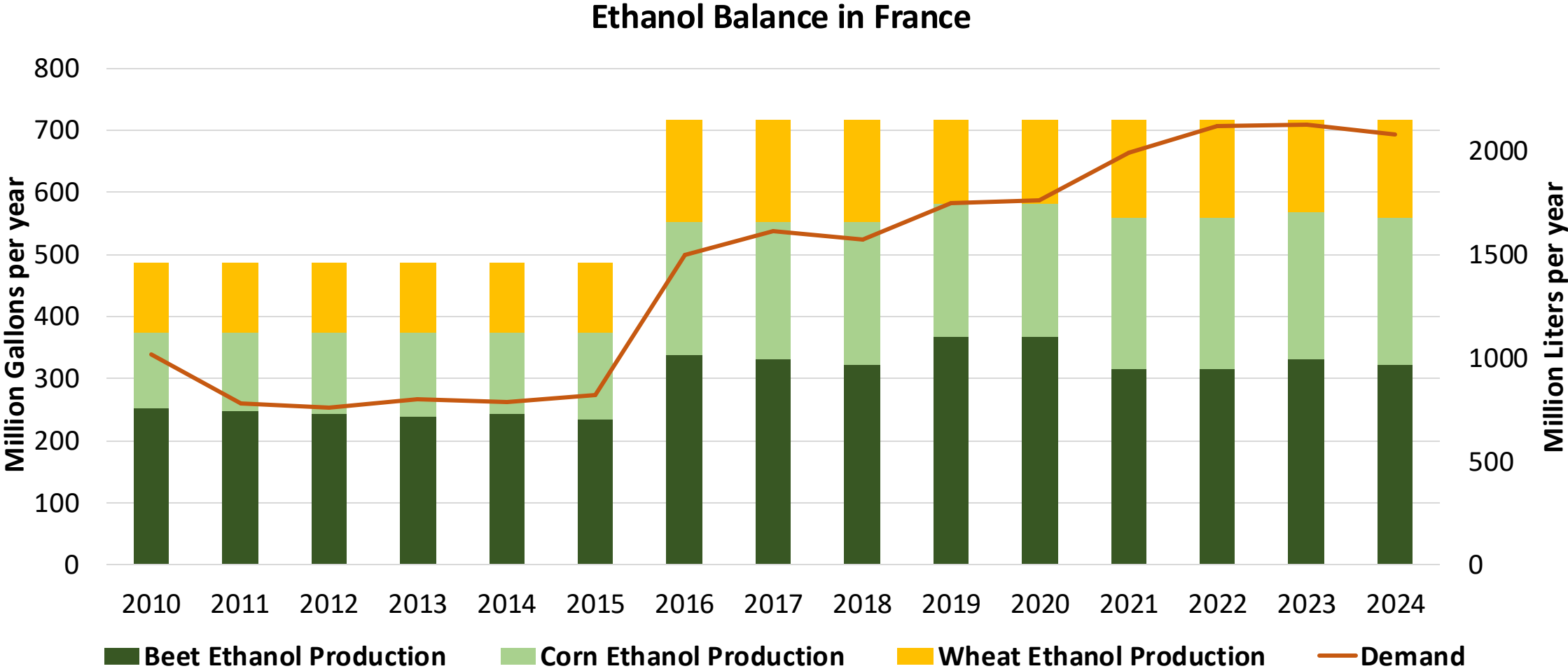
Gasoline Imports in France



Gasoline Exports in France



Ethanol Balance in France



Source: Eurostat

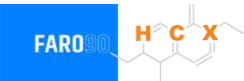
Gasoline Quality in France

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Countries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<= 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	1.2							
Bioethanol in Gasoline Sales (%cal)	8.6							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commission

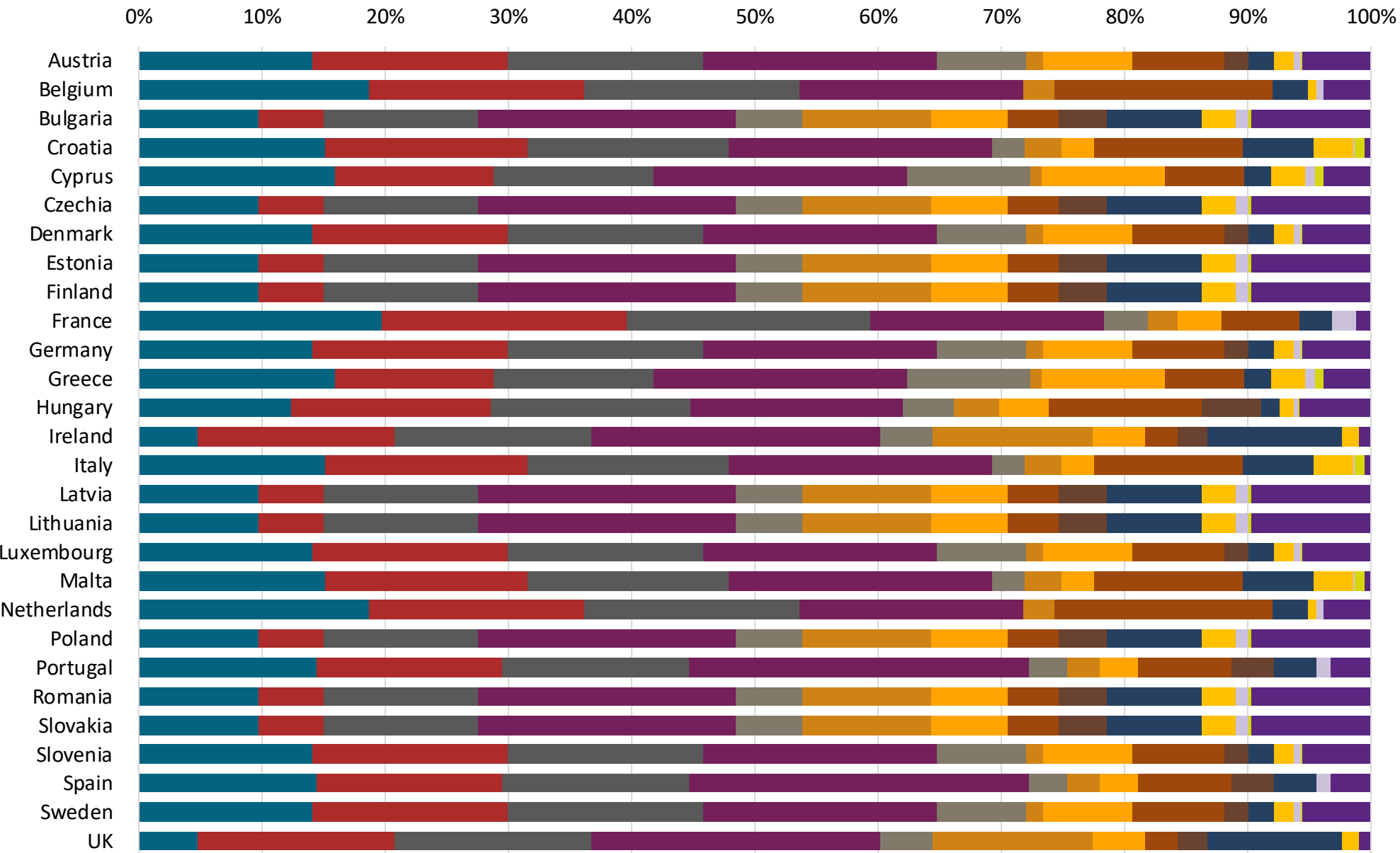
Gasoline Component Blending in Europe



Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics

Source: Faro90



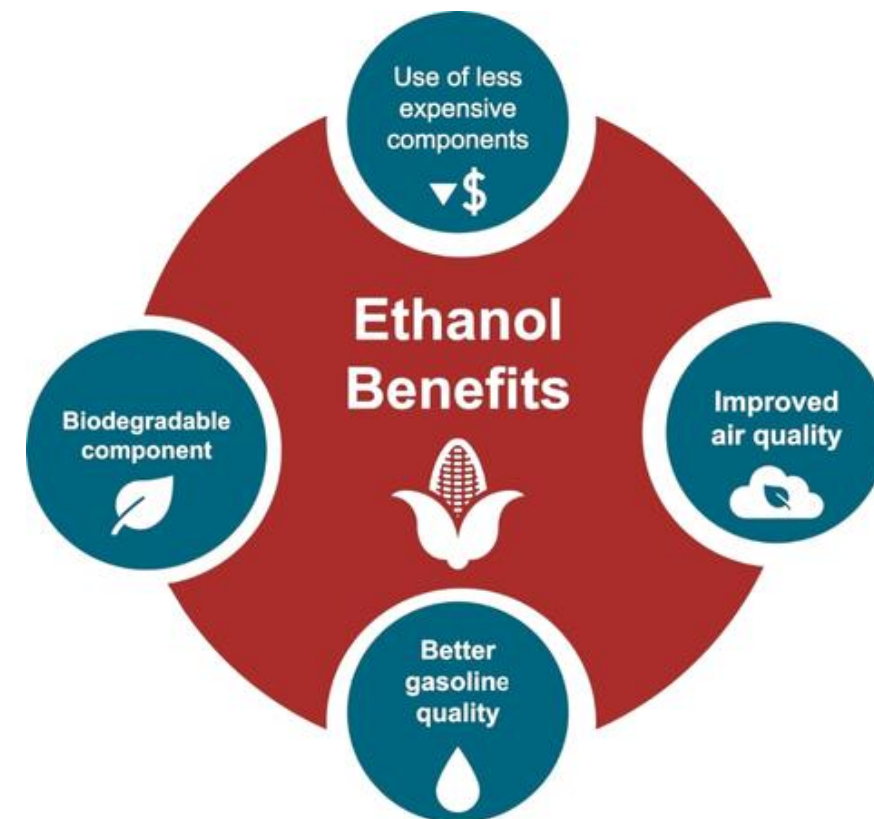
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

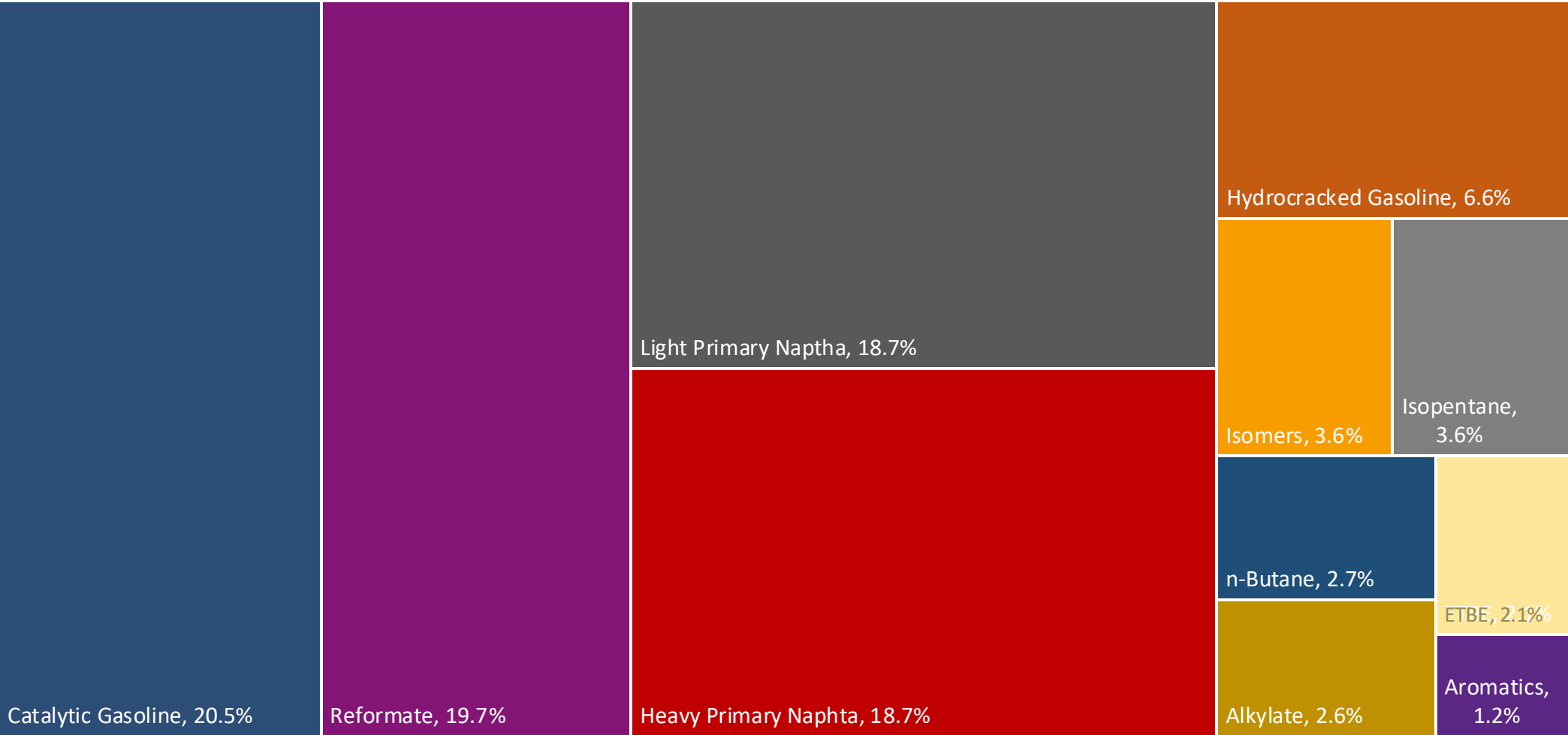
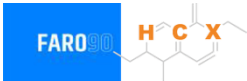
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

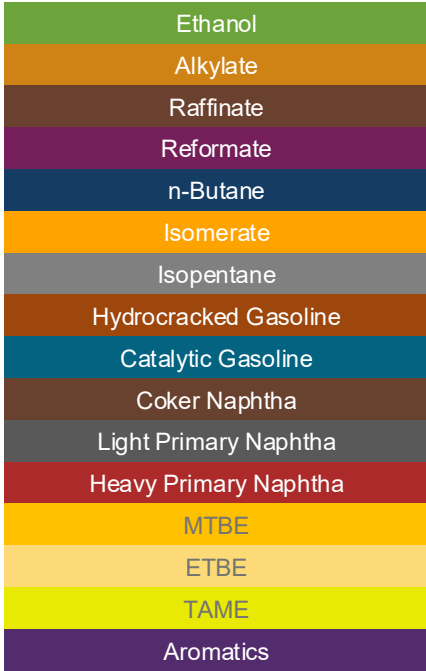
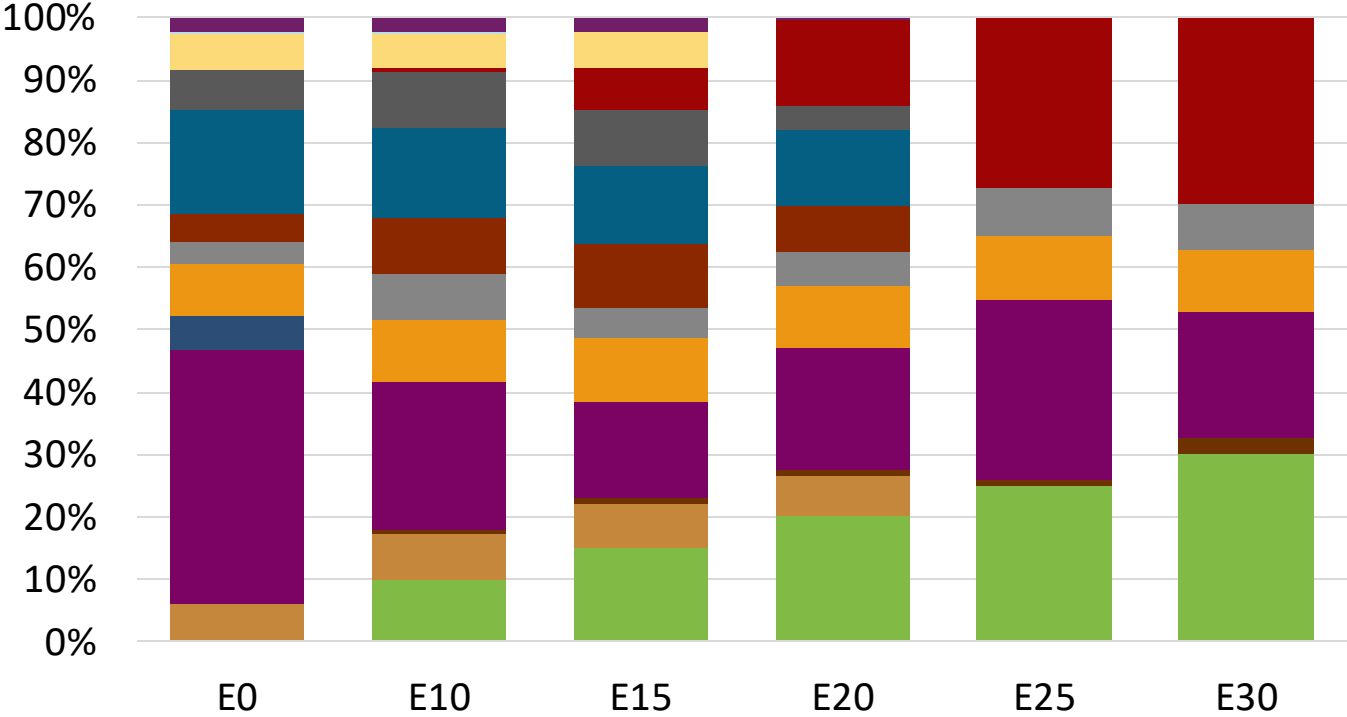
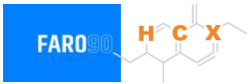
The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



France Available Blending Components



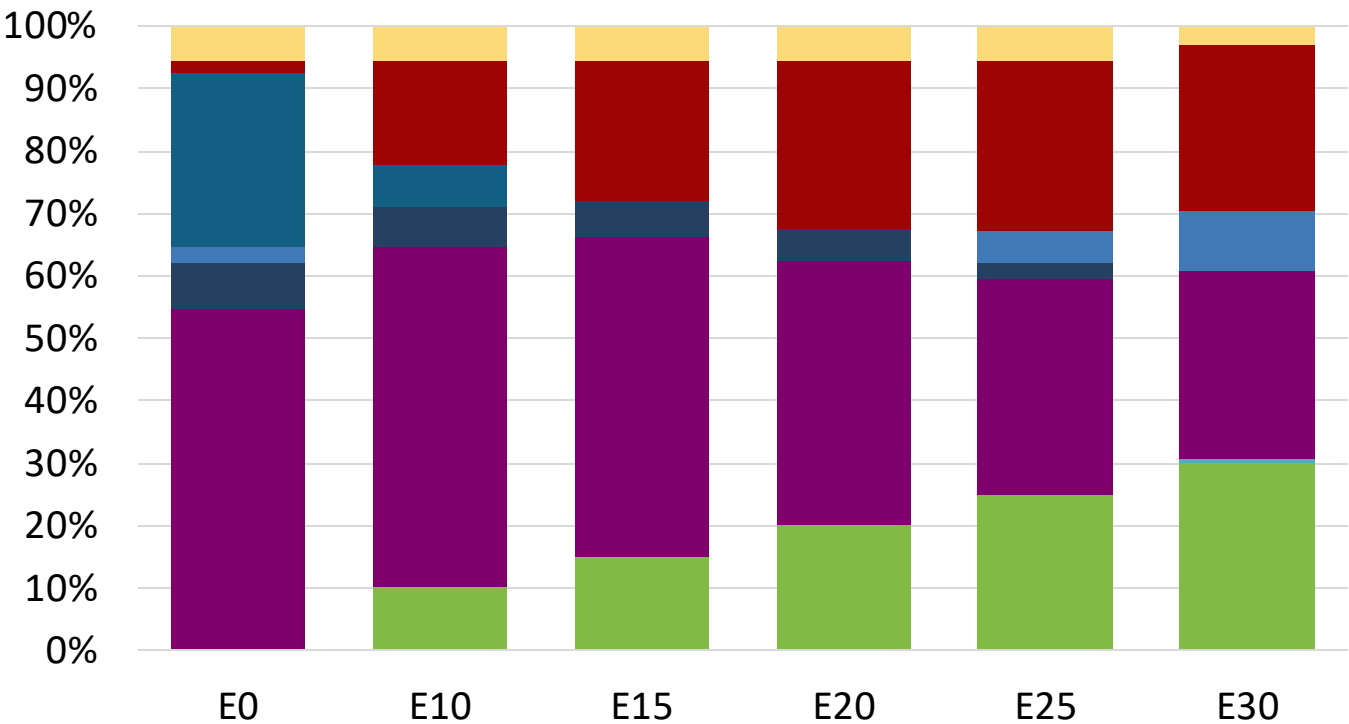
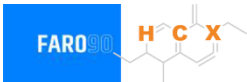
France – Gasoline 95 – Constant Octane



Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.2	95.7
Price (US\$/gal)	\$ 2.268	\$ 2.158	\$ 2.079	\$ 1.996	\$ 1.882	\$ 1.810

France - Gasoline 98 - Constant Octane

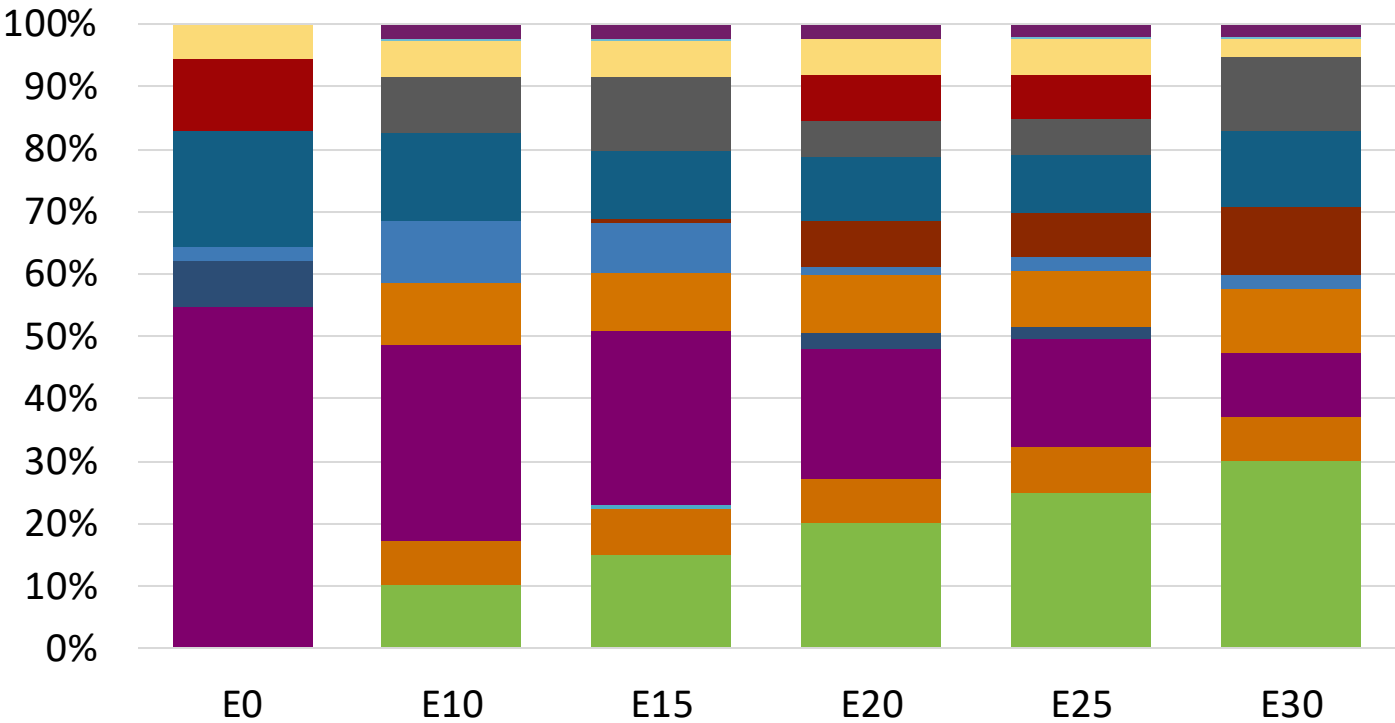
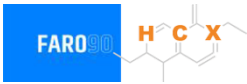


Average CIF 2024 Prices

Octane (RON)	98.0					
Price (US\$/gal)	\$ 2.346	\$ 2.222	\$ 2.151	\$ 2.076	\$ 1.998	\$ 1.923

Source: Faro90

France – Gasoline 95 – Increased Octane

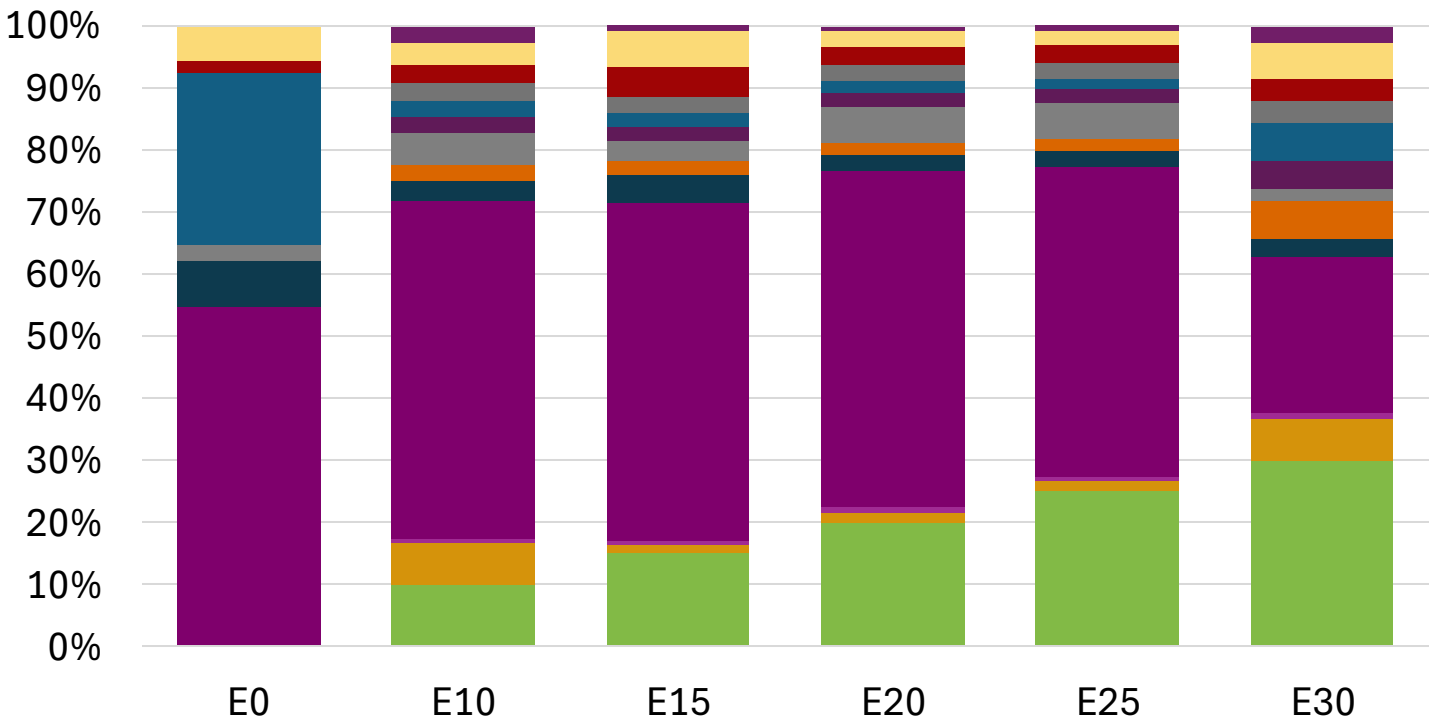


Average CIF 2024 Prices

Octane (RON)	95.0	96.5	97.9	99.4	100.8	101.7
Price (US\$/gal)	\$ 2.212	\$ 2.212	\$ 2.212	\$ 2.212	\$ 2.212	\$ 2.212

Source: Faro90

France – Gasoline 98 – Increased Octane



Average CIF 2024
Prices

Octane (RON)		98.0	99.3	101.4	102.4	103.7	104.6
Price (US\$/gal)	\$	2.346	\$ 2.346	\$ 2.346	\$ 2.346	\$ 2.346	\$ 2.346

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

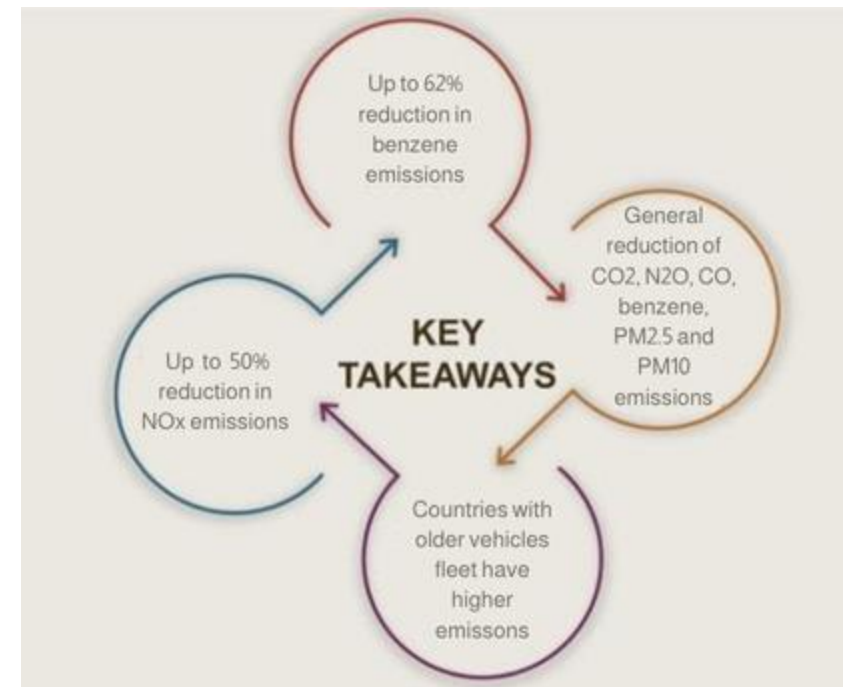
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

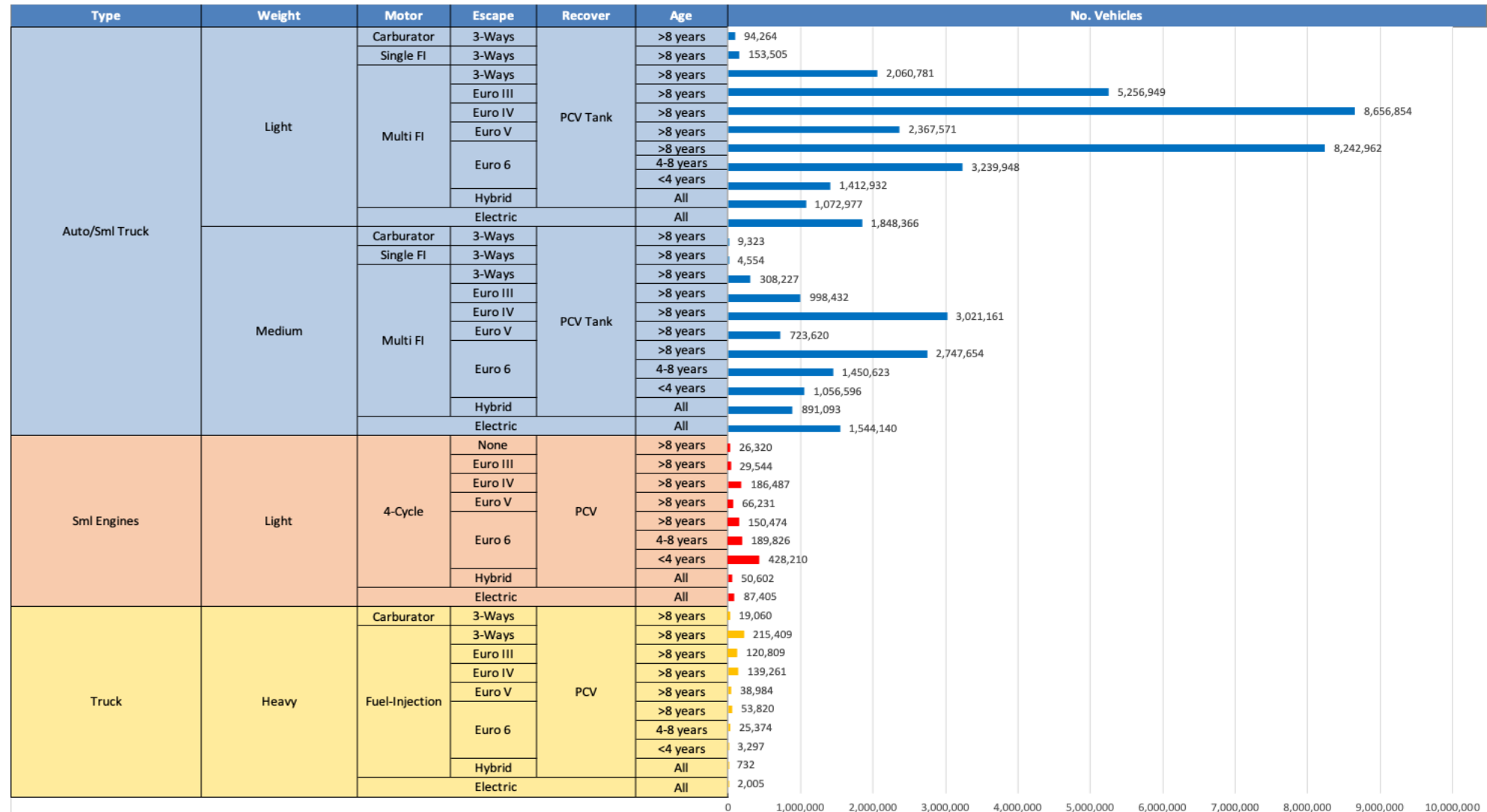
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

*Source: Bureau of transportation statistics.

Main Results



Gasoline Vehicle Fleet - France

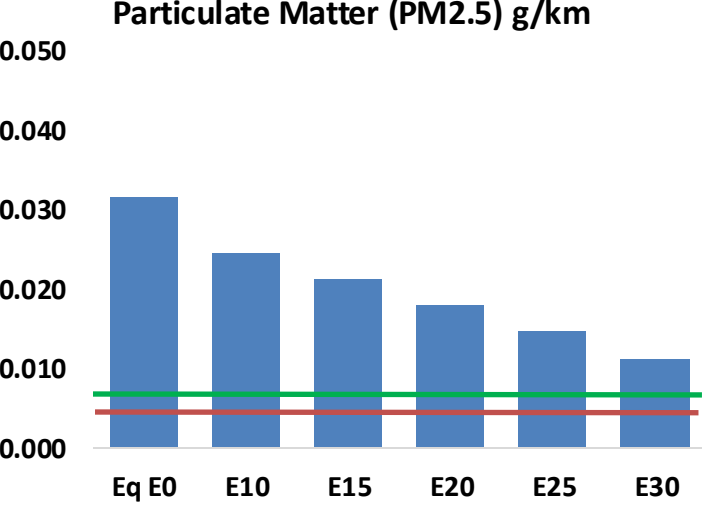
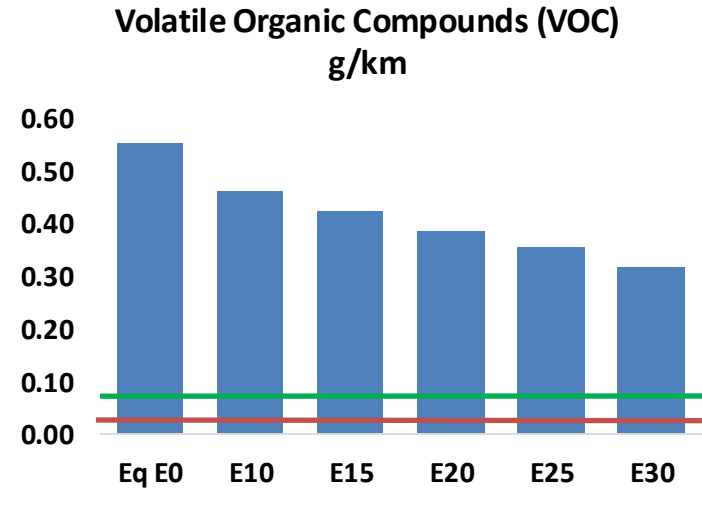
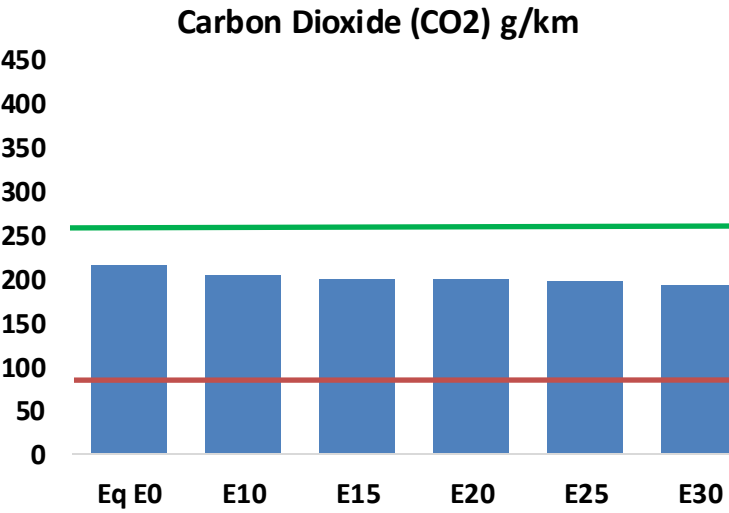
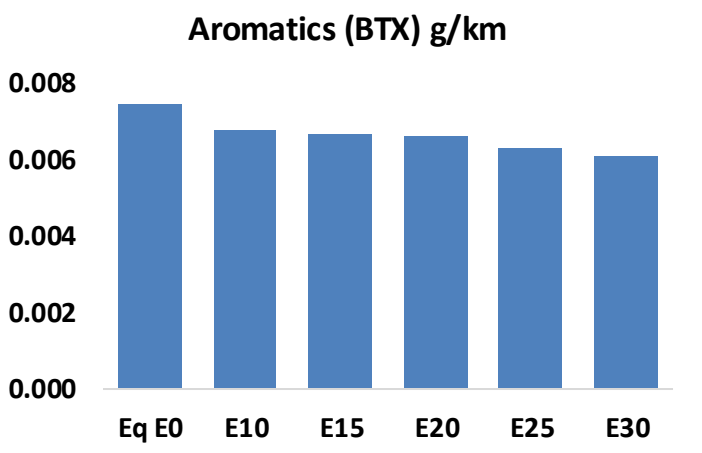
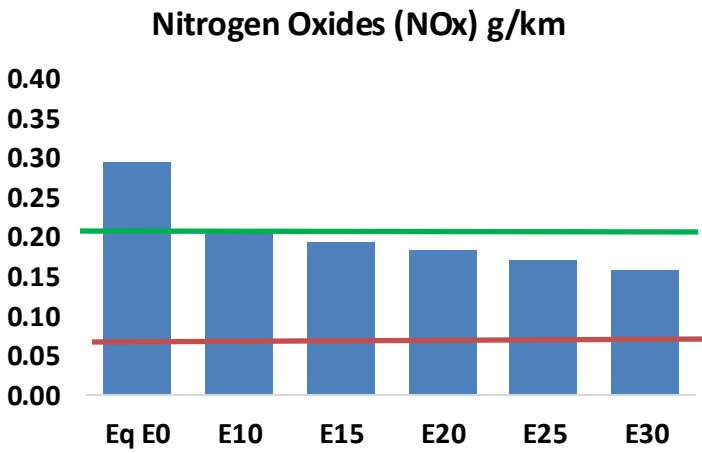
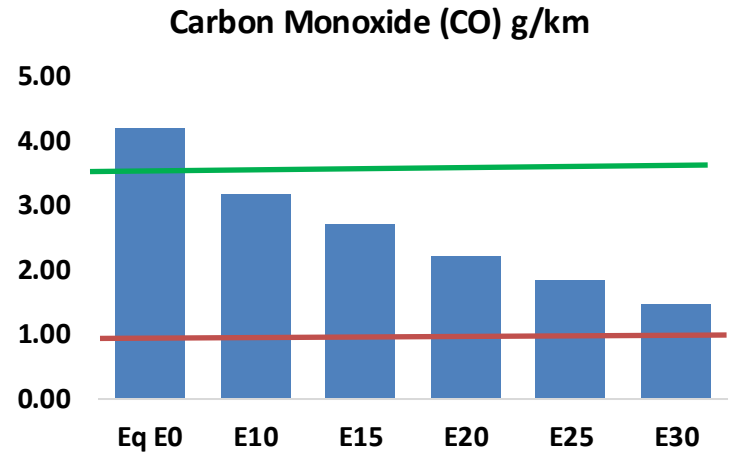
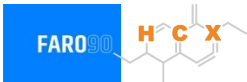


Vehicle Fleet: 48,964,974
Gasoline Vehicle Fleet: 24,096,952

Source: Eurostat, ACEA

France – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



France – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	2,605,439	2,284,795	1,964,150	1,677,912	1,379,054	1,145,988	912,581	-25%	-47%
CO2	133,470,121	130,038,032	126,796,615	124,247,335	122,985,852	122,244,413	119,991,097	-5%	-8%
VOC	343,295	314,878	286,460	262,778	238,764	220,244	195,723	-17%	-30%
NOx	182,003	136,502	127,402	120,076	113,242	105,707	97,736	-30%	-38%
SOx	1,536	1,420	1,304	1,206	1,112	1,010	902	-15%	-28%
NH3	47,099	46,633	46,166	46,385	46,907	47,274	47,579	-2%	0%
N2O	1,983	1,972	1,964	1,974	2,014	2,037	2,060	-1%	2%
CH4	70,260	70,260	70,260	71,314	72,859	74,194	75,459	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	2,334	2,199	2,065	1,963	1,864	1,790	1,686	-12%	-20%
Acetaldehyde	5,934	6,810	9,993	15,218	20,733	23,691	27,994	68%	249%
Formaldehyde	20,829	20,251	23,606	27,719	29,040	32,126	34,973	13%	39%
Benzene	4,604	4,424	4,192	4,126	4,092	3,913	3,787	-9%	-11%
PM2.5	8,142	7,229	6,317	5,486	4,658	3,780	2,864	-22%	-43%
PM10	11,339	10,069	8,798	7,640	6,488	5,265	3,989	-22%	-43%

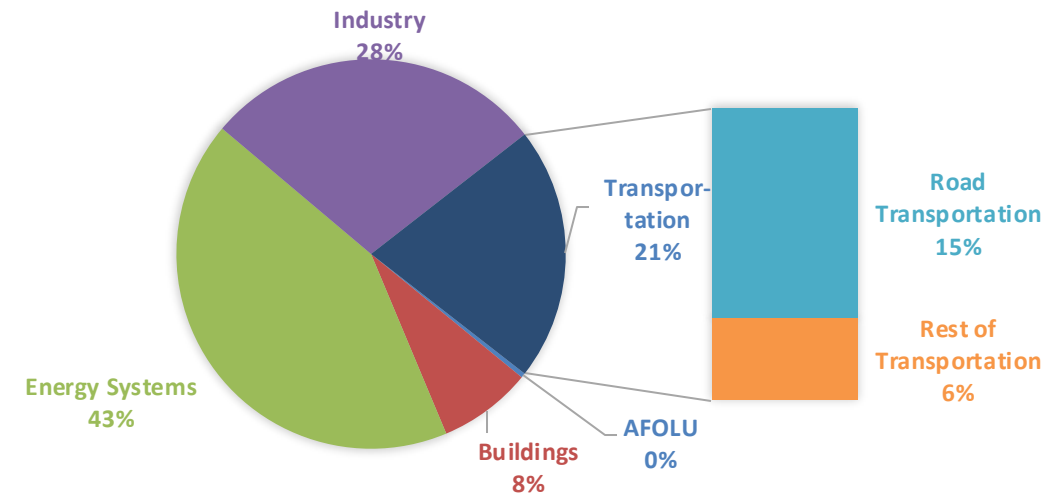
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

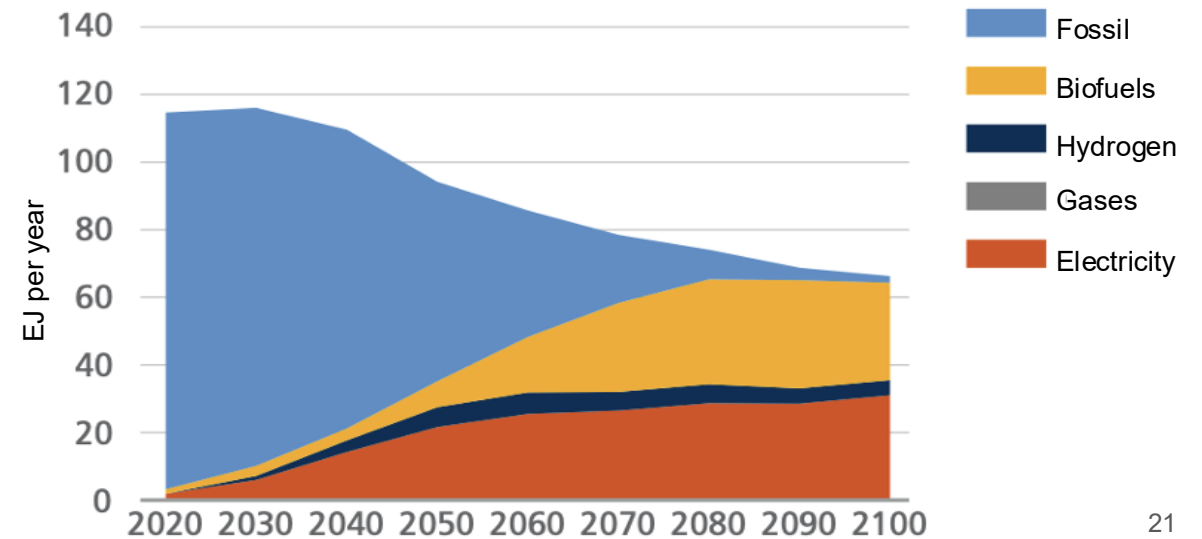
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

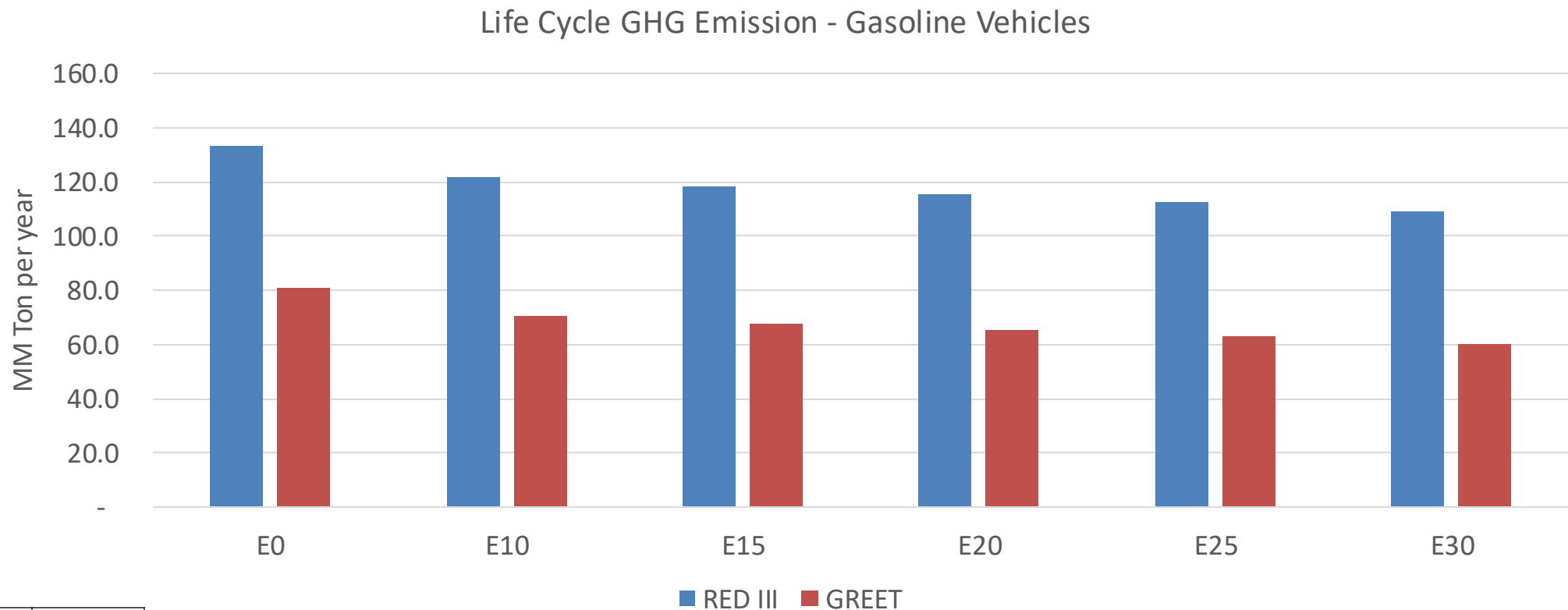
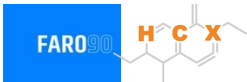
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



France – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	331.1
Target @ 2035 (MMT/year)	186.4
Delta	144.7

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.9%	16.4%	19.2%	22.0%	25.6%
RED II/III	0.0%	8.8%	11.4%	13.6%	15.6%	18.2%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	7.2%	9.2%	10.8%	12.3%	14.3%
RED II/III	0.0%	8.1%	10.5%	12.5%	14.4%	16.8%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90